Multivariate Regime-Conditional Simulator

Simulated World Project.

Part 1.3: Synthetic Worlds. Factors, Return SImulation

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.neighbors import KernelDensity
from hmmlearn.hmm import GaussianHMM
from scipy import stats
from sklearn.linear_model import LinearRegression
from statsmodels.stats.diagnostic import acorr_ljungbox
```

Load data

Real world Data.

In [2]:
```
# All features includes 'return', 'fwd_return', 'regime'
data_2 = pd.read_csv("project_DATA/_vet_features_df.csv", index_col="Date", parse_dates=True)
vet_features_df = data_2.copy(deep=True)
print("======== Full set of features. vet_features_df:")
vet_features_df
```

```
======== Full set of features. vet_features_df:
```

Out[2]:

Date	Volume	VIX_indx_close	term_spread	T10Y3M	T10Y2Y_yield_curve	DGS10_dif	T10Y2Y_dif	DTB3_dif	news_sentiment	is_GDP_release_day	...	ADX_norm	
2000-04-12	1175900000	28.980000	6.40	0.30	1	0.08	0.04	0.00	0.25	0	...	0.196604	-0.0
2000-04-13	1032000000	29.400000	6.40	0.29	1	-0.03	-0.03	-0.02	0.23	0	...	0.203679	-0.0
2000-04-14	1279700000	33.490002	6.27	0.20	1	-0.09	0.04	0.00	0.22	0	...	0.226684	-0.0
2000-04-17	1204700000	28.950001	6.33	0.36	1	0.16	0.10	0.00	0.20	0	...	0.248046	0.0
2000-04-18	1109400000	26.120001	6.33	0.40	1	0.04	0.04	0.00	0.20	0	...	0.253507	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	
2025-12-23	3820560000	14.000000	3.48	0.62	0	0.01	-0.03	0.00	-0.02	1	...	0.128592	0.0
2025-12-24	1798270000	13.470000	3.47	0.55	0	-0.03	-0.02	0.04	-0.00	0	...	0.134678	0.0
2025-12-26	2586550000	13.600000	3.46	0.58	0	-0.01	0.00	-0.04	-0.01	0	...	0.141582	-0.0
2025-12-29	3541750000	14.200000	3.45	0.55	0	-0.02	-0.01	0.01	0.04	0	...	0.140478	-0.0
2025-12-30	3309930000	14.330000	3.45	0.59	0	0.02	0.02	-0.02	0.03	0	...	0.139453	-0.0

6468 rows × 68 columns

In [3]:
```
# Factors metrics sorted by pro_score_penalized. Correlated
data_3 = pd.read_csv("project_DATA/_pro_score_df_metrics_factors.csv")
factor_metrics_sorted = data_3.copy(deep=True)
```

In [4]:
```
# Factors-Metrics based on selected factors
factor_metrics = factor_metrics_sorted.T
headers = factor_metrics.iloc[0].values
factor_metrics.columns = headers
factor_metrics = factor_metrics.drop(index='factors')
print("======== factor_metrics:")
factor_metrics
```

```
======== factor_metrics:
```

Out[4]:

	RSI_7_norm	RSI_14_norm	price_vs_EMA20	price_vs_EMA10	price_vs_SMA50	BB_position	high_close_ratio	price_vs_SMA20	VIX_indx_close	ROC_10	...
pro_score	2.288127	1.87097	1.622433	1.411629	1.368139	1.362533	1.244761	1.104468	1.068365	1.035916	...
robust_score_3	2.037878	1.643392	1.435035	1.345246	1.138493	1.337038	1.289867	1.027222	0.889143	1.027313	...
ic_all	-0.056772	-0.048241	-0.044146	-0.05007	-0.03262	-0.047682	0.059749	-0.040647	0.02941	-0.041829	...
ic_calm	-0.08233	-0.079805	-0.066322	-0.064675	-0.058681	-0.07143	0.079496	-0.063137	0.068082	-0.049404	...
ic_stress	-0.049543	-0.041717	-0.044277	-0.050893	-0.03663	-0.035603	0.085627	-0.036647	0.066273	-0.035603	...
regime_gap	0.032787	0.038087	0.022045	0.013782	0.022052	0.035827	0.006132	0.026491	0.001809	0.013801	...
vol_sensitivity	0.034777	0.041967	0.032597	0.021256	0.036608	0.042795	0.007601	0.037935	0.037213	0.033933	...
stability_metric	1.391535	1.455437	1.378795	1.155797	1.563794	1.161985	0.949948	1.264372	1.502979	1.08059	...
bootstrap_IC_mean	-0.057212	-0.047761	-0.044276	-0.050432	-0.032899	-0.048197	0.060025	-0.040491	0.027781	-0.042236	...
bootstrap_IC_interval	0.039066	0.042299	0.042541	0.04333	0.045188	0.041887	0.044206	0.049839	0.046961	0.044427	...
ci_contains_zero	False	False	False	False	False	False	False	False	False	False	...
gap_gt_ic	False	False	False	False	False	False	False	False	False	False	...
pro_score_penal	2.288127	1.87097	1.622433	1.411629	1.368139	1.362533	1.244761	1.104468	1.068365	1.035916	...
score_delta	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...

14 rows × 24 columns

In [5]:
```
factor_metrics.shape
```

Out[5]:
```
(14, 24)
```

In [6]:
```
# Read selected_factors
print("======== Selected Factors:")
selected_factors_list = factor_metrics.columns.to_list()
selected_factors_list
```

```
======== Selected Factors:
```

Out[6]:
```
['RSI_7_norm',
 'RSI_14_norm',
 'price_vs_EMA20',
 'price_vs_EMA10',
 'price_vs_SMA50',
 'BB_position',
 'high_close_ratio',
 'price_vs_SMA20',
 'VIX_indx_close',
 'ROC_10',
 'close_position',
 'body',
 'ROC_20',
 'Low_return',
 'news_sentiment',
 'Open_return',
 'upper_wick',
 'DGS10_dif',
 'gap',
 'T10Y3M',
 'kin_f3_cpi_t10y3m',
 'BB_width',
 'DTB3_dif',
 'gap_filled']
```

In [7]:
```
# the Markov transition matrix learned by GaussianHMM on real data
hmm_matrix = np.load('project_DATA/HMM_real_world_transition_matrix.npy')
trans_mat = hmm_matrix.copy()

print("======== HMM_trans_mat:")
print(trans_mat)
print(trans_mat.shape)
```

```
======== HMM_trans_mat:
[[9.79538250e-01 2.04617504e-02 1.06263579e-90]
 [1.80061367e-02 9.72568467e-01 9.42539597e-03]
 [1.87665236e-58 2.16112139e-02 9.78388786e-01]]
(3, 3)
```

In [8]:
```
# For research perpouse matrix. More aggresive stress regime
#trans_mat_research = [[0.97, 0.025, 0.005], [0.009, 0.97, 0.021], [0.001, 0.029, 0.97]]
```

Real world HMM Regime Statistics

In [9]:
```
# HMM Regime Statistics
def hmm_regime_stat(df, col):
    sub_df = df[[col]].copy(deep=True)
```

```
    sub_df['run_id'] = (sub_df[col] != sub_df[col].shift()).cumsum()
    run_stats = sub_df.groupby([col, 'run_id']).size().reset_index(name='length')

    print("Section details:")
    print(run_stats)

    print("\nSummary:")
    summary = run_stats.groupby(col).agg(
        num_sections=('run_id', 'nunique'),
        total_count=('length', 'sum')
    )
    print(summary)

    return run_stats
```

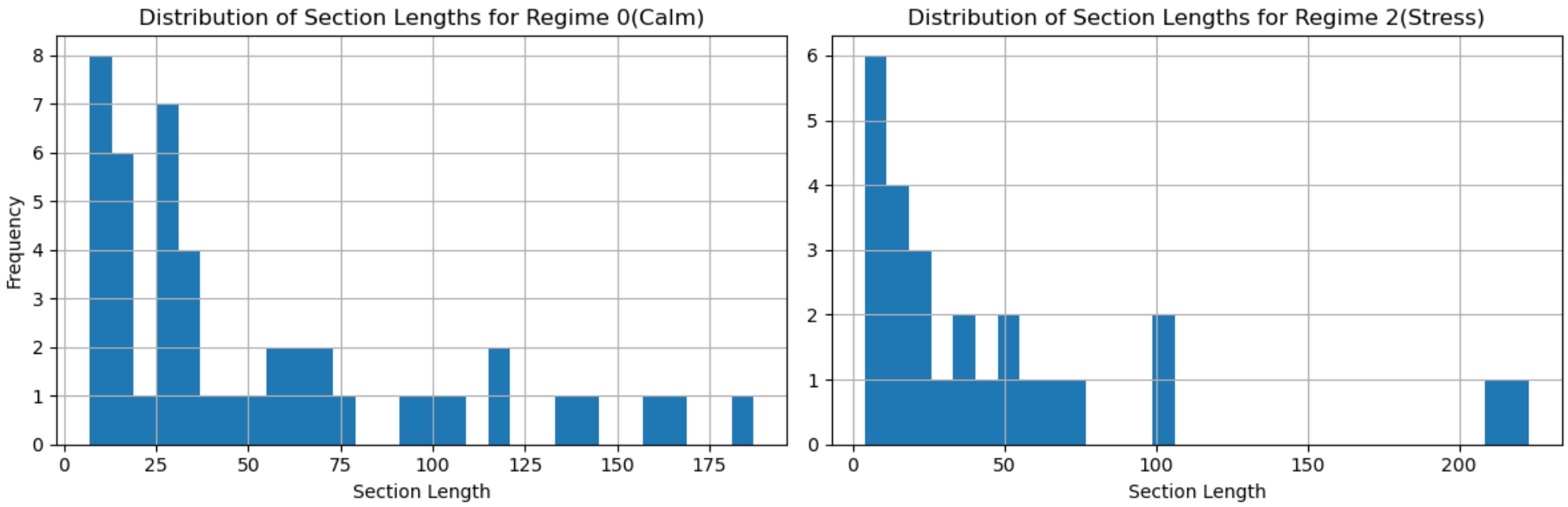
In [10]:
```
# Real Stat
run_stat_real = hmm_regime_stat(vet_features_df, col='hmm_regime')
```

```
Section details:
     hmm_regime  run_id  length
0             0      13      37
1             0      15      17
2             0      17      48
3             0      19     164
4             0      21     116
..          ...     ...     ...
138           2     119       6
139           2     121      21
140           2     123      59
141           2     125      75
142           2     139      31

[143 rows x 3 columns]

Summary:
            num_sections  total_count
hmm_regime
0                     46         2431
1                     71         2773
2                     26         1264
```

In [11]:
```
# Calm(regime 0) vs Stress(regime 2) Distributions Vizualization
def calm_stress_regime_viz(stats_df, col):
    fig, axes = plt.subplots(1, 2, figsize=(12, 4))

    # Plot for regime 0
    stats_df[stats_df[col] == 0]['length'].hist(ax=axes[0], bins=30)
    axes[0].set_title('Distribution of Section Lengths for Regime 0(Calm)')
```

```
    axes[0].set_xlabel('Section Length')
    axes[0].set_ylabel('Frequency')

    # Plot for regime 1
    stats_df[stats_df[col] == 2]['length'].hist(ax=axes[1], bins=30)
    axes[1].set_title('Distribution of Section Lengths for Regime 2(Stress)')
    axes[1].set_xlabel('Section Length')

    # Plot for regime 2


    plt.tight_layout()
    plt.show()
```

In [12]:
```
# Check the regimes destribution on Real world
calm_stress_regime_viz(run_stat_real, col='hmm_regime')
```



Simulate Regime

Simulate the world, Copula path

In [13]:
```
# 1. Simulate regimes. Use the real world HMM probability matrix
def simulate_regimes(T, trans_mat, init_state=0):
    states = [init_state]

    for _ in range(T-1):
        current = states[-1]
        next_state = np.random.choice(
            len(trans_mat),
            p=trans_mat[current]
        )
        states.append(next_state)
        col = ['hmm_regime']

    return pd.DataFrame(data=np.array(states), columns=col)
```

In [14]:
```
# number of observations
T=len(vet_features_df)
```

Simulate regimes

In [22]:
```
simulate_regimes_df = simulate_regimes(T, trans_mat, init_state=0)
simulate_regimes_df['hmm_regime'].value_counts(dropna=False)
```

Out[22]:
```
hmm_regime
0    2867
1    2233
2    1368
Name: count, dtype: int64
```

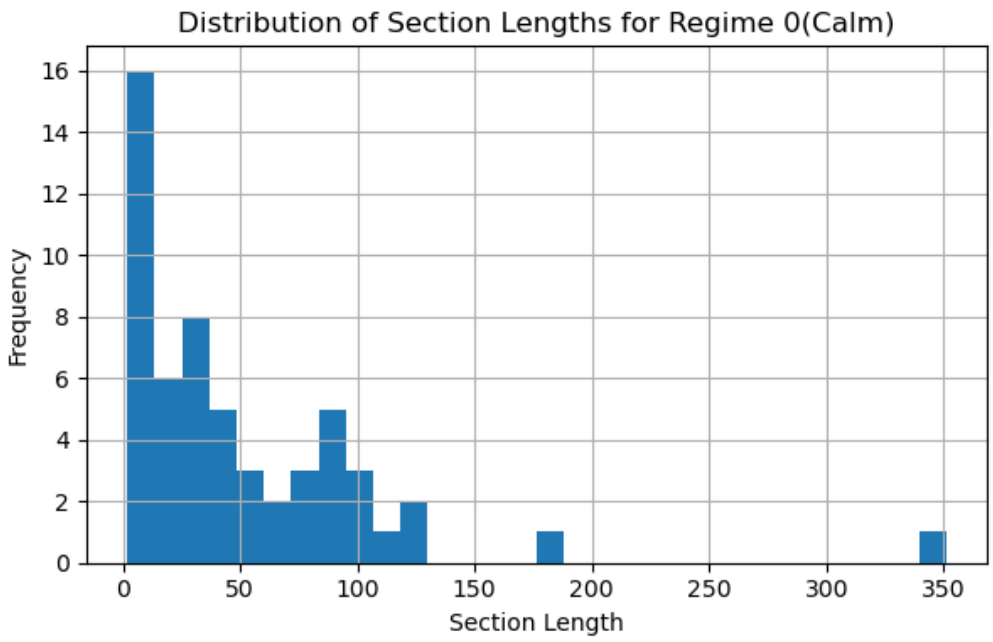
Simulated world HMM Regime Statistics

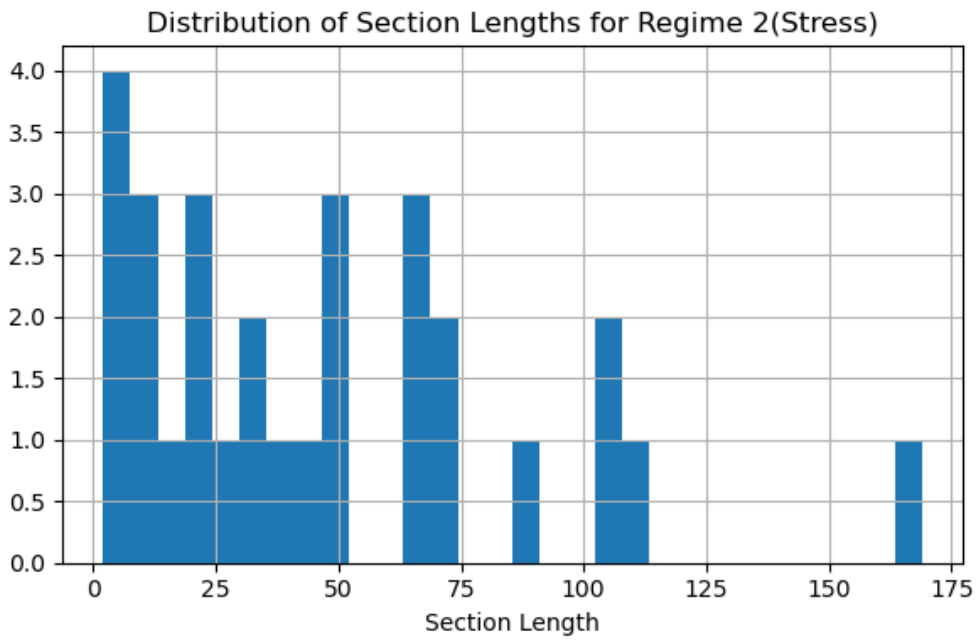
In [23]:
```
# Simulated Stat
run_stat_sim = hmm_regime_stat(simulate_regimes_df, col='hmm_regime')
```

```
Section details:
     hmm_regime  run_id  length
0             0       1       7
1             0       3      12
2             0      11       6
3             0      19      87
4             0      21      35
..          ...     ...     ...
164           2     133       7
165           2     147      86
166           2     155      12
167           2     159      19
168           2     163      51

[169 rows x 3 columns]

Summary:
            num_sections  total_count
hmm_regime
0                     56         2867
1                     84         2233
2                     29         1368
```

In [24]:

```
# Check the regimes destribution on Simulated world
calm_stress_regime_viz(run_stat_sim, col='hmm_regime')
```





Gaussian Copula Simulator

```
'''
Any joint distribution can be decomposed into two separate pieces –
the marginal distributions of each variable individually,
and a copula that describes the dependence structure between them.
'''
```

Fit Multivariate per regime. Goal: to define the "parameters" of selected factors to simulate them

In [25]:
```
count_regime_0 = (simulate_regimes_df['hmm_regime'] == 0).sum()
count_regime_1 = (simulate_regimes_df['hmm_regime'] == 1).sum()
count_regime_2 = (simulate_regimes_df['hmm_regime'] == 2).sum()
count_regime_0, count_regime_1, count_regime_2
```

Out[25]:
```
(np.int64(2867), np.int64(2233), np.int64(1368))
```

In [26]:
```
# ================ Step 1 – Rank-Gaussian transform
def rank_gauss(x):
    r = stats.rankdata(x)/(len(x)+1)      # Convert to percentiles (0 to 1). Original => Uniform
    return stats.norm.ppf(r)              # # Map percentiles to Gaussian quantiles. Uniform => Gaussian
```

In [27]:
```
# ============== Step 2 — Fit copula per regime.
# Learning the Dependence structure separately for each market regime.

def fit_copula(df_in, features, regime_col):

    df = df_in.copy(deep=True)

    models = {}

    for r in sorted(df[regime_col].unique()):
        sub = df[df[regime_col]==r][features].dropna()
        print(f"len of sub{r}['data']: {len(sub)}")
        # Transforms each column from original distribution to standard normal.
        Z = np.column_stack([
            rank_gauss(sub[c]) for c in features
        ])

        Sigma = np.cov(Z.T) # np.cov() expects variables in rows, observations in columns

        models[r] = {
            "cov": Sigma,
            "data": sub
        }

    return models
```

In [28]:
```
# Get Copula destribution models
models = fit_copula(vet_features_df, selected_factors_list, regime_col="hmm_regime")
```

```
len of sub0['data']: 2431
len of sub1['data']: 2773
len of sub2['data']: 1264
```

In [29]:
```
len(selected_factors_list)
```

Out[29]:
```
24
```

In [30]:
```
# Factors sampling simulation function
def sample_copula(models, features, regime, n):
    '''
    Generating Synthetic Data
    '''
    m = models[regime]

    # Generates new samples with the same correlation structure
    Z = np.random.multivariate_normal(
        np.zeros(len(features)),
```

```
        m["cov"],
        size=n
    )

    U = stats.norm.cdf(Z)
    X = np.zeros_like(U)

    '''
    **Inverse ECDF** — uses the uniform values as quantile lookup indices into the original sorted real-world data.
    So if `U=0.9`, it picks the 90th percentile value from the actual observed distribution of that feature.
    This is what makes it non-parametric.
    '''
    # inverse ECDF. Restores realistic non-normal distributions (fat tails, skewness)
    for j,c in enumerate(features):
        sorted_vals = np.sort(m["data"][c])
        idx = (U[:,j]*(len(sorted_vals)-1)).astype(int)

        # Takes data from real set but This works because of Sklar's Theorem:
        '''
        The Statistical Principle
        The dependency structure is separate from the marginal distributions
        We can "break" the original rows and "recombine" values
        As long as we preserve:
        The correlation structure (via Gaussian copula)
        The individual distributions (via lookup from sorted values)
        '''
        X[:,j] = sorted_vals[idx]

    return pd.DataFrame(data=X, columns=features)
```

In [31]:

```
# Generate Factors Sample DF based on regime
copula_factors_sim_0 = sample_copula(models, selected_factors_list, regime=0, n=count_regime_0)
copula_factors_sim_1 = sample_copula(models, selected_factors_list, regime=1, n=count_regime_1)
copula_factors_sim_2 = sample_copula(models, selected_factors_list, regime=2, n=count_regime_2)
# Copy
simulated_F_R0 = copula_factors_sim_0.copy(deep=True)
simulated_F_R1 = copula_factors_sim_1.copy(deep=True)
simulated_F_R2 = copula_factors_sim_2.copy(deep=True)

simulated_F_R0.shape, simulated_F_R1.shape, simulated_F_R2.shape
```

Out[31]:

```
((2867, 24), (2233, 24), (1368, 24))
```

Concut simuated_regime_df(regime column only) + simulated_F_R0, simulated_F_R1, simulated_F_R2

```
row by row based on regime mark 0,1,2
```

In [32]:
```
# Create a mapping array for each regime
regime_indices = {
    0: np.arange(len(simulated_F_R0)),
    1: np.arange(len(simulated_F_R1)),
    2: np.arange(len(simulated_F_R2))
}

# Create a counter for each regime
regime_positions = {0: 0, 1: 0, 2: 0}
regime_column = 'hmm_regime'

# Pre-allocate list for efficiency
rows_list = []

for regime in simulate_regimes_df[regime_column]:
    regime = int(regime)
    pos = regime_positions[regime]

    # Get the specific row
    if regime == 0:
        row = simulated_F_R0.iloc[[pos]]
    elif regime == 1:
        row = simulated_F_R1.iloc[[pos]]
    else:  # regime == 2
        row = simulated_F_R2.iloc[[pos]]

    rows_list.append(row)
    regime_positions[regime] += 1

# Combine
df_first_sim = pd.concat(rows_list, ignore_index=True)
df_first_sim['hmm_regime'] = simulate_regimes_df[regime_column].values
```

First Factor's Simulated Set

In [33]:
```
# First simulated set of selected factors + 'hmm_regime'
df_first_sim.shape
```

Out[33]: (6468, 25)

Final Set with Cliped Simulated Factor's

In [34]:
```
# Clip sim_factors
def clip_to_empirical_by_regime(df_sim_sim, df_real_real, factors):
    """
    Clip simulated values to empirical quantiles based on regime.
```

```
    Parameters:
    df_sim: DataFrame with simulated data (will be modified)
    df_real: DataFrame with real data containing 'hmm_regime' column
    factors: list of column names to clip
    """
    df_sim = df_sim_sim.copy(deep=True)
    df_real = df_real_real.copy(deep=True)

    # Get unique regimes
    regimes = df_real['hmm_regime'].unique()

    for regime in regimes:
        # Select data for this regime
        mask = df_real['hmm_regime'] == regime
        regime_data = df_real.loc[mask]

        if len(regime_data) == 0:
            continue

     # Set quantiles based on regime
        if regime == 0:
            lo_q, hi_q = 0.01, 0.99
        elif regime == 2:
            lo_q, hi_q = 0.001, 0.999
        else:
            # Default quantiles for other regimes
            lo_q, hi_q = 0.01, 0.99

        # Clip each feature for this regime
        for col in factors:
            if col in regime_data.columns:
                lo = regime_data[col].quantile(lo_q)
                hi = regime_data[col].quantile(hi_q)

                # Apply clipping only to rows in this regime
                regime_mask_sim = df_sim['hmm_regime'] == regime
                df_sim.loc[regime_mask_sim, col] = df_sim.loc[regime_mask_sim, col].clip(lo, hi)

    return df_sim
```

In [35]:
```
# Simulate clipted_factors, no returns simulated
df_clipped_sim = clip_to_empirical_by_regime(df_first_sim, vet_features_df, selected_factors_list)
```

Check the factors destribution on real world vs simulated

In [36]:
```
# Random factors set to check destribution
factors_to_check = ['VIX_indx_close', 'high_close_ratio', 'DTB3_dif', 'Open_return', 'kin_f3_cpi_t10y3m', 'T10Y3M']
```

```python
# Real world Selected factors destrib
print(f"==================")
print(f"||     Real     ||")
print(f"==================")

vet_features_df[factors_to_check].describe()
```

```
==================
||     Real     ||
==================
```

Out[36]:

	VIX_indx_close	high_close_ratio	DTB3_dif	Open_return	kin_f3_cpi_t10y3m	T10Y3M
count	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000
mean	19.799405	1.006087	-0.000252	0.000300	-1.429431	1.439487
std	8.401821	0.007971	0.043878	0.011369	1.300672	1.296867
min	9.140000	1.000000	-0.810000	-0.087119	-4.271095	-1.730000
25%	13.960000	1.001192	-0.010000	-0.004421	-2.472500	0.350000
50%	17.690001	1.003283	0.000000	0.000756	-1.550000	1.560000
75%	23.132499	1.007976	0.010000	0.005730	-0.350000	2.480000
max	82.690002	1.104767	0.740000	0.106712	2.017783	3.870000

In [37]:

```python
# First Simulated
print(f"==================")
print(f"||  Simulated_1 ||")
print(f"==================")

df_first_sim[factors_to_check].describe()
```

```
==================
||  Simulated_1 ||
==================
```

Out[37]:

	VIX_indx_close	high_close_ratio	DTB3_dif	Open_return	kin_f3_cpi_t10y3m	T10Y3M
count	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000
mean	19.461020	1.005834	-0.000057	0.000481	-1.399596	1.404283
std	8.524329	0.007495	0.044210	0.011455	1.277890	1.276002
min	9.150000	1.000000	-0.810000	-0.087119	-3.820000	-1.730000
25%	13.500000	1.001175	-0.010000	-0.003982	-2.432500	0.330000
50%	16.700001	1.003184	0.000000	0.000842	-1.520000	1.530000
75%	23.162500	1.007664	0.010000	0.005798	-0.330000	2.450000
max	80.860001	1.095567	0.610000	0.106513	1.730000	3.810000

In [38]:

```
# After bouds Standartization and cliped by regimes
print(f"==================")
print(f"|| Simulated_2  ||")
print(f"==================")

df_clipped_sim[factors_to_check].describe()
```

```
==================
|| Simulated_2  ||
==================
```

Out[38]:

	VIX_indx_close	high_close_ratio	DTB3_dif	Open_return	kin_f3_cpi_t10y3m	T10Y3M
count	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000	6468.000000
mean	19.461241	1.005811	-0.000121	0.000472	-1.399692	1.404479
std	8.519191	0.007442	0.040859	0.011369	1.276073	1.274202
min	9.636000	1.000000	-0.548960	-0.083953	-3.820000	-1.472800
25%	13.530000	1.001175	-0.010000	-0.003982	-2.432500	0.330000
50%	16.700001	1.003184	0.000000	0.000842	-1.520000	1.530000
75%	23.162500	1.007664	0.010000	0.005798	-0.330000	2.450000
max	80.649600	1.094833	0.575810	0.099045	1.482800	3.720000

simulate RETURNS

```
Simulate Returns from factors (β-model) add temporal memory (AR structure)
- regime-conditioned β model
- regime AR(1)
- regime volatility
- fat tails

Regime simulated srtuctural model:
r(t) = mu(r) + b(r)T(F(t) - mu(F,r)) + phi(r) * r(t-1) + eps(t,r)
eps(t,r) ~ t_v(0,sigma(r))
```

In [39]:
```python
# AR() returns on Real world
'''
Calculates the AR(1) autocorrelation coefficient (phi) -
is the lag-1 autocorrelation coefficient
for returns within each HMM regime separately
'''
def regime_ar1(df_rw, regime):
    df = df_rw.copy(deep=True)

    # Selects only rows belonging to the target regime (0, 1, or 2).
    mask = df["hmm_regime"] == regime

    r_t = df["return"]
    r_lag = r_t.shift(1) # return from previous period
    # Each row now has r_t (today) and r_lag (yesterday).

    # Key filter — both today AND yesterday must be in the same regime.
    # This avoids correlating a return from one regime with a return
    # from a completely different regime context.
    valid = mask & mask.shift(1)

    return np.corrcoef(r_lag[valid], r_t[valid])[0,1]
```

In [40]:
```python
# phi by regime
phi_0 = regime_ar1(vet_features_df, regime=0)   # calm
phi_1 = regime_ar1(vet_features_df, regime=1)   # transition
phi_2 = regime_ar1(vet_features_df, regime=2)   # stress
```

In [41]:
```python
# Real world phi dict
phi_dict_real = {
    0: phi_0,   # calm
    1: phi_1,   # transition
    2: phi_2    # stress
}
```

In [42]:
```
phi_dict_real
```

Out[42]:
```
{0: np.float64(-0.06953743497372676),
 1: np.float64(-0.03864943565307201),
 2: np.float64(-0.14768316509818616)}
```

In [43]:
```
# Estimate on REAL data
def calibrate_regime_model(df_rw, features,
                           return_col="fwd_return",
                           regime_col="hmm_regime"):
    '''
    This function calibrates a regime-switching model on Real Data.
    It estimates separate parameters for each regime (state) in the data,
    including regression coefficients, means, and volatility.

    Parameters:
    'df_rw': Input DataFrame containing the data
    'features': List of column names to use as predictor variables (factors)
    'return_col': Name of the target variable column (default: "fwd_return")
    'regime_col': Name of the column identifying which regime each row belongs to (default: "hmm_regime")

    Output:
    Returns a tuple of 4 dictionaries:
    "betas": Regression coefficients for each regime (slope factor)
    "factor_means": Mean values of features for each regime
    'return_means': Mean return for each regime
    'sigma_dict': Residual volatility for each regime
    '''

    df = df_rw.copy(deep=True)
    betas = {}            # Dictionary to store regression coefficients
    factor_means = {}
    return_means = {}
    #phi_dict = {}
    sigma_dict = {}

    for r in sorted(df[regime_col].unique()):

        sub = df[df[regime_col] == r].copy()

        # -------------------------
        # AR(1)
        # -------------------------
        #phi = sub[return_col].autocorr(lag=1)
        #phi_dict[r] = phi

        # -------------------------
        # Center X and y
```

```python
        # -------------------------
        X = sub[features]
        y = sub[return_col]

        mu_F = X.mean().values   # numpy.ndarray
        mu_r = y.mean()          # scalar

        X_centered = X - mu_F
        y_centered = y - mu_r

        # -------------------------
        # Regression
        # -------------------------
        reg = LinearRegression()
        reg.fit(X_centered, y_centered)

        beta = reg.coef_
        '''
        'beta' is a NumPy array containing the regression coefficients (slopes)
        for each feature in the model.
        Each coefficient tells you how much the target variable (return) changes
        when that specific feature changes by one unit, holding all other features constant.
        '''

        # -------------------------
        # Residual volatility
        # -------------------------
        residuals = y_centered - reg.predict(X_centered) # Prediction errors (epsilon)
        # Positive = model under-predicted
        # Negative = model over-predicted

        print(f"Regime {r}, Mean residuals: {np.mean(residuals)}")

        sigma_r = residuals.std()    # Standard deviation of residuals (of these errors)
        '''
        'sigma_r' is the standard deviation of the residuals -
        the part of returns that model cannot explain using the features.
        It's a measure of the model's uncertainty or the "noise" in each regime.

        f.e.
        calm regime 0:
        sigma_r = 0.008 --> 0.8% volatility => Model is quite precise - errors are small and tight
        stress regime 2:
        sigma_r = 0.045 --> 4.5% volatility => Model is less precise - errors are larger and more spread out
        '''

        # -------------------------
        # Store
```

```
        # -------------------------
        betas[r] = beta
        factor_means[r] = mu_F
        return_means[r] = mu_r
        sigma_dict[r] = sigma_r

        print(f"Regime {r}: sigma_r = {sigma_dict[r]:.4f}")
        print(f"In Regime {r}:")
        print(f"  68% of prediction errors +/- {sigma_dict[r] * 100:.2f}%")
        print(f"  95% of prediction errors +/- {2 * sigma_dict[r] * 100:.2f}%")
        print(f"  Maximum typical error: +/-   {3 * sigma_dict[r] * 100:.2f}%")

    return betas, factor_means, return_means, sigma_dict
```

In [44]:
```
betas, factor_means, return_means, sigma_dict = calibrate_regime_model(
    vet_features_df, selected_factors_list,
    return_col="fwd_return", regime_col="hmm_regime"
)
```

```
Regime 0, Mean residuals: 1.1417349080883963e-20
Regime 0: sigma_r = 0.0059
In Regime 0:
  68% of prediction errors +/- 0.59%
  95% of prediction errors +/- 1.18%
  Maximum typical error: +/-   1.76%
Regime 1, Mean residuals: -4.003689234133273e-20
Regime 1: sigma_r = 0.0099
In Regime 1:
  68% of prediction errors +/- 0.99%
  95% of prediction errors +/- 1.98%
  Maximum typical error: +/-   2.97%
Regime 2, Mean residuals: 0.0
Regime 2: sigma_r = 0.0212
In Regime 2:
  68% of prediction errors +/- 2.12%
  95% of prediction errors +/- 4.24%
  Maximum typical error: +/-   6.36%
```

Smulated Returns Regeme Engine

In [45]:
```
def simulate_returns_regime_engine(df_sim_sim,
                                   features,
                                   betas,
                                   factor_means,
                                   return_means,
                                   phi_dict,
                                   sigma_dict,
```

```
                                regime_col="hmm_regime",
                                df_t=8):

    df_sim = df_sim_sim.copy(deep=True)
    sim_r = np.zeros(len(df_sim))

    #--------------------------------#
    # initialise conditional variance #
    h = None
    omega = None
    prev_r = None

    # Starting values (GARCH-lite parameters):
    alpha = 0.02    # shock sensitivity
    beta_g = 0.90   # variance persistence
    #--------------------------------#

    for t in range(len(df_sim)):

        row = df_sim.iloc[t]
        r = int(row[regime_col])

        #--------------------------------#
        # When the regime changes (calm->stress), sigma_r jumps.
        # We need to reset h at each regime transition to avoid a discontinuity:
        # Re-initialise h when regime changes
        if r != prev_r:
            h = sigma_dict[r]**2
            omega = sigma_dict[r]**2 * (1 - alpha - beta_g) # = sigma_r**2 × 0.08
            prev_r = r
        #--------------------------------#

        beta = betas[r]
        mu_F = factor_means[r]
        mu_r = return_means[r]
        phi = phi_dict[r]
        sigma_r = sigma_dict[r]

        F = row[features].values

        # centered signal
        signal = np.dot(beta, F - mu_F)

        # regime-specific fat-tailed noise. Scaled by today's conditional std
        #eps = np.random.standard_t(df=df_t) * sigma_r
        eps = np.random.standard_t(df=df_t) * np.sqrt(h)

        if t == 0:
```

```
            sim_r[t] = mu_r + signal + eps
        else:
            sim_r[t] = mu_r + signal + phi * sim_r[t-1] + eps

        #---------------------------------#
        # Update conditional variance (GARCH-lite)
        # Return today = (volatility today) × (Student-t fat-tailed noise)
        # Volatility tomorrow = Long-term average + α × (Return today)^2 + β × (Volatility today)
        h = omega + alpha * eps**2 + beta_g * h
        #---------------------------------#

    df_sim["sim_return"] = sim_r
    return df_sim
```

In [46]:
```
# DF clipped_sim with 'return' simulated
df_clipped_sim_new = simulate_returns_regime_engine(
    df_clipped_sim, selected_factors_list, betas,
    factor_means, return_means, phi_dict_real,
    sigma_dict, regime_col="hmm_regime", df_t=8
)
```

In [47]:
```
#================================ VIZ CLOSE SIM vs REGIME SIM ========================================#
```

In [48]:
```
# Define your date ranges
def viz_sim_close_vs_regime_1500(combined_df):

    date_ranges = [
        (0, 1499),
        (1500, 2999),
        (3000, 4499),
        (4500, 5999),
        (6000, 6499)
    ]

    start_date = 0
    end_date = 6468

    # Create subplots
    n_plots = len(date_ranges)
    fig, axes = plt.subplots(n_plots, 1, figsize=(15, 5*n_plots))

    colors = {0: 'blue', 1: 'green', 2: 'red'}

    for idx, (start_date, end_date) in enumerate(date_ranges):
        # Filter data for this date range
        mask = (combined_df.index >= start_date) & (combined_df.index <= end_date)
        df_range = combined_df[mask]
```

```
        if len(df_range) > 0:
            ax = axes[idx]

            # Plot each regime
            for regime in df_range['hmm_regime'].unique():
                regime_mask = df_range['hmm_regime'] == regime

                if regime == 0:
                    regime_name = "calm"
                elif regime == 1:
                    regime_name = "transition"
                else:
                    regime_name = "stress"

                ax.scatter(df_range.index[regime_mask],
                           df_range['sim_close'][regime_mask],
                           c=colors.get(regime, 'gray'),
                           label=f'Regime {regime_name}',
                           s=10, alpha=0.7)

            ax.set_title(f'Simulated Period: {start_date} to {end_date}', fontsize=12)
            ax.set_ylabel('Simulated Close Price', fontsize=10)
            ax.grid(True, alpha=0.3)
            ax.tick_params(axis='x', rotation=45)
            ax.legend()

    plt.tight_layout()
    plt.show()
```

In [49]:
```
# Generate realistic price data
regimes_sim = df_clipped_sim_new["hmm_regime"].copy(deep=True)
returns_sim = df_clipped_sim_new["sim_return"].copy(deep=True)
```

In [50]:
```
# Generate realistic price data
start_price = 1000
close_sim = start_price * np.exp(np.cumsum(returns_sim))
# Convert to data frame
regimes_sim_df = pd.DataFrame(regimes_sim)
close_sim_df = pd.DataFrame(close_sim)
```

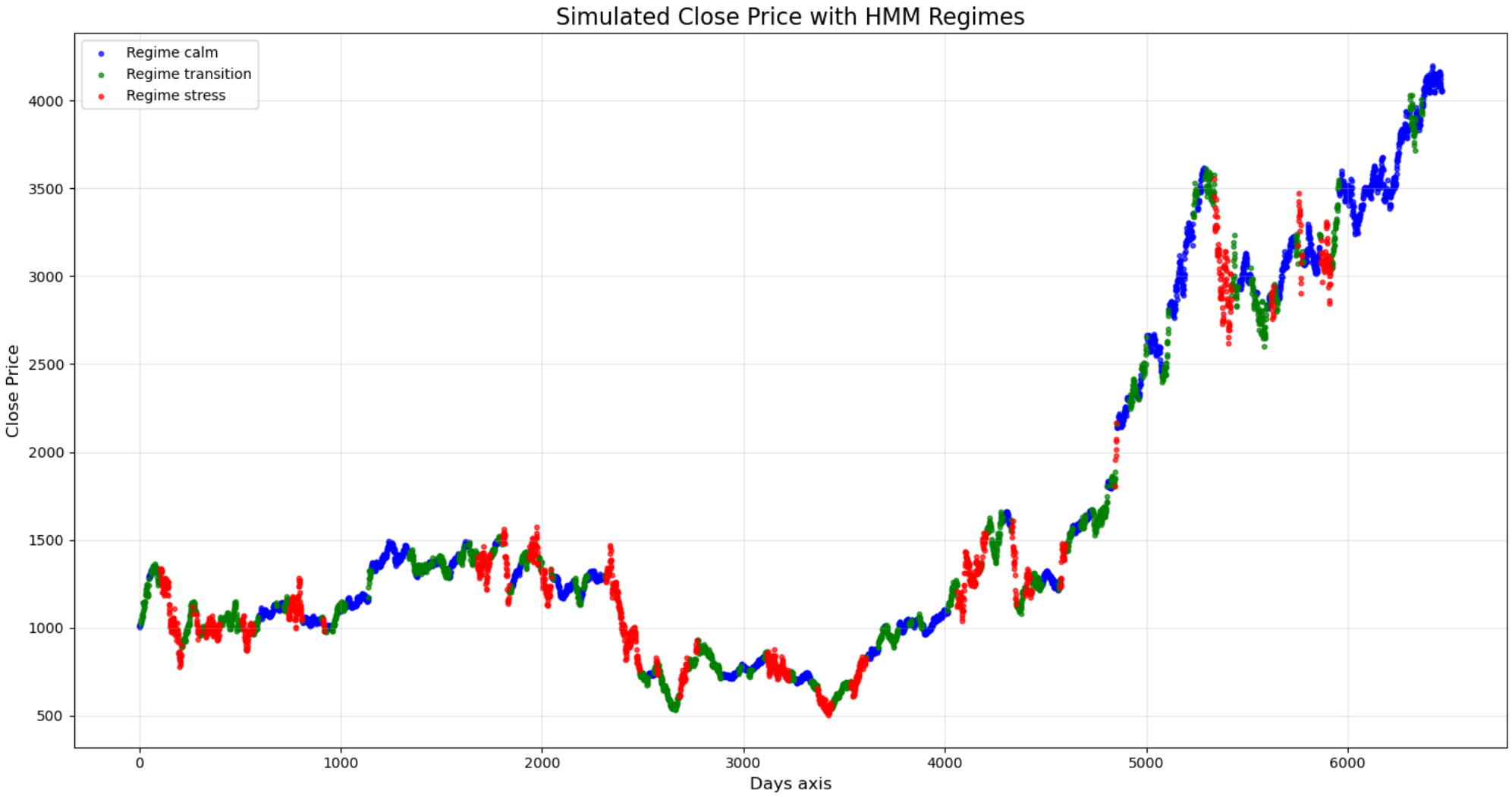
In [51]:
```
# Align the data (make sure dates match)
combined_sim_df = pd.concat([close_sim_df, regimes_sim_df], axis=1).dropna()
combined_sim_df.rename(columns={'sim_return': 'sim_close'}, inplace=True)
```

In [52]:
```
combined_sim_df['sim_close'].min(), combined_sim_df['sim_close'].max()
```

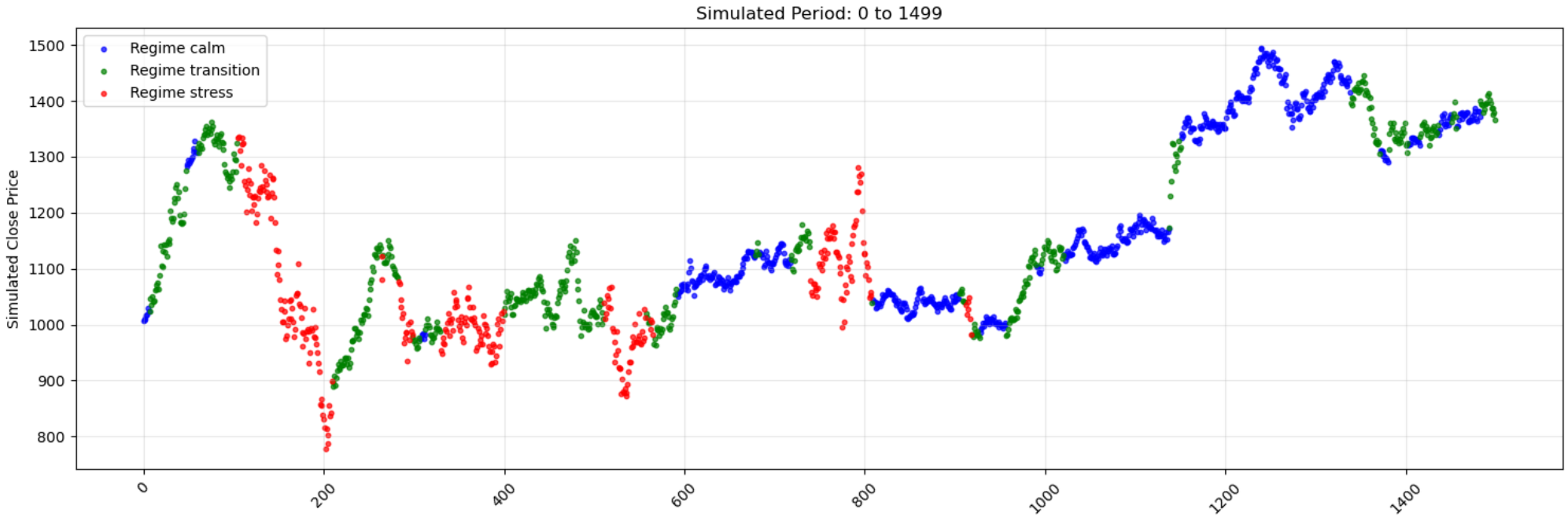
Out[52]: (np.float64(505.47705192041093), np.float64(4196.8715756107895))

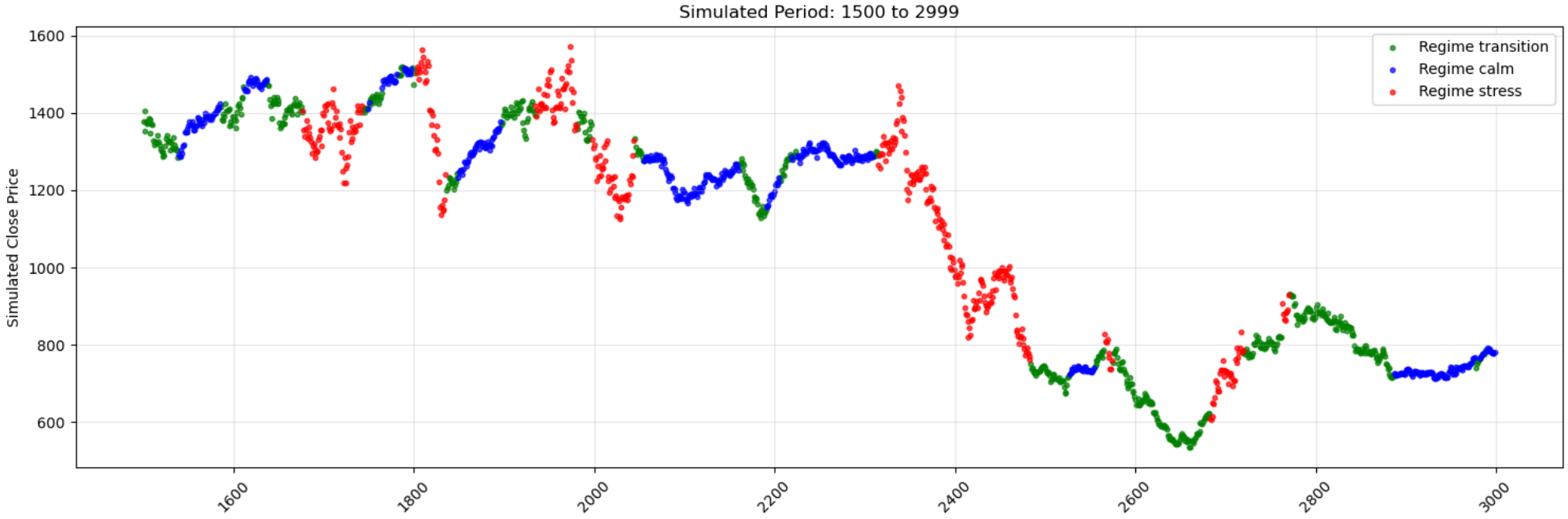
In [53]:
```
def all_data_viz(combined_df):
    # Create the plot
    plt.figure(figsize=(15, 8))

    # Scatter plot with colors based on regime
    colors = {0: 'blue', 1: 'green', 2: 'red'}
    for regime in combined_df['hmm_regime'].unique():
        mask = combined_df['hmm_regime'] == regime

        if regime == 0:
            regime_name = "calm"
        elif regime == 1:
            regime_name = "transition"
        else:
            regime_name = "stress"


        plt.scatter(combined_df.index[mask],
                    combined_df['sim_close'][mask],
                    c=colors.get(regime, 'gray'),
                    label=f'Regime {regime_name}',
                    s=10, alpha=0.7)

    plt.title('Simulated Close Price with HMM Regimes', fontsize=16)
    plt.xlabel('Days axis', fontsize=12)
    plt.ylabel('Close Price', fontsize=12)
    plt.legend()
    plt.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.show()
```

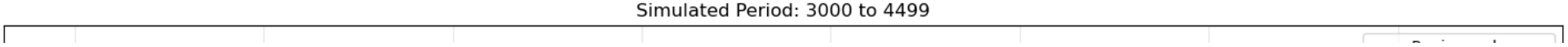
In [54]:
```
all_data_viz(combined_sim_df)
```



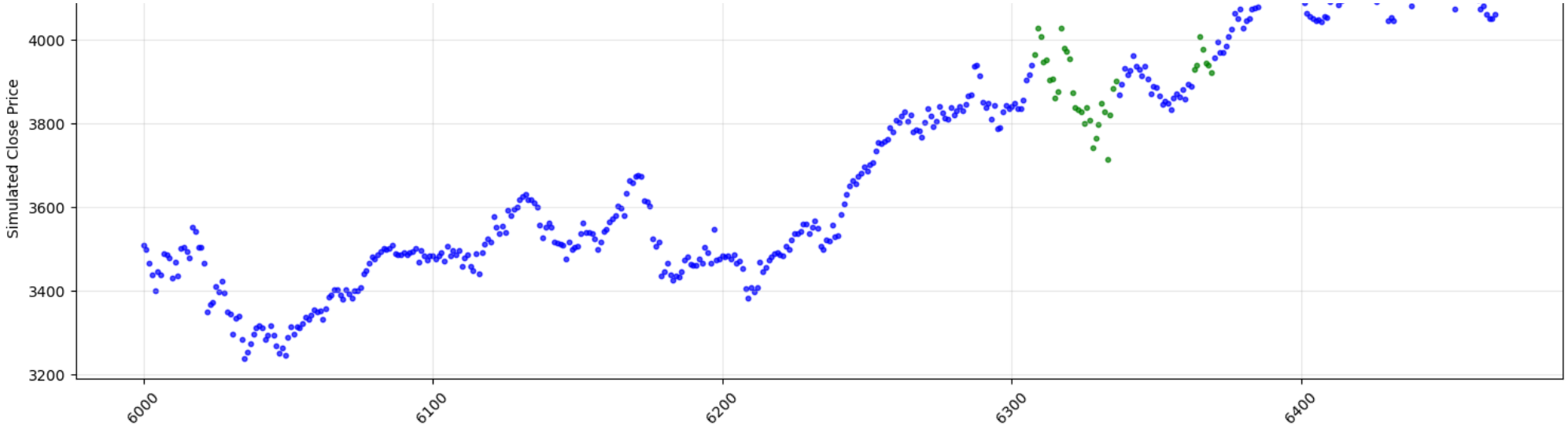
In [55]:
```
# Sliced dayes
viz_sim_close_vs_regime_1500(combined_sim_df)
```


Simulated Period: 0 to 1499
Regime calm
Regime transition
Regime stress
Simulated Close Price
1500
1400
1300
1200
1100
1000
900
800
0
200
400
600
800
1000
1200
1400


Simulated Period: 1500 to 2999
Regime transition
Regime calm
Regime stress
Simulated Close Price
1600
1400
1200
1000
800
600
1600
1800
2000
2200
2400
2600
2800
3000


Simulated Period: 3000 to 4499

Regime calm
Regime transition
Regime stress
Simulated Close Price
1600
1400
1200
1000
800
600
3000
3200
3400
3600
3800
4000
4200
4400
Simulated Period: 4500 to 5999
Regime calm
Regime transition
Regime stress
Simulated Close Price
3500
3000
2500
2000
1500
4600
4800
5000
5200
5400
5600
5800
6000
Simulated Period: 6000 to 6499
Regime calm
Regime transition
4200



Level Checks Sim vs Real

Destribution

In [56]:
```
# Real data 'high_close_ratio'
vet_features_df[selected_factors_list + ["hmm_regime"] + ['fwd_return']]["high_close_ratio"].describe()
```

Out[56]:
```
count    6468.000000
mean        1.006087
std         0.007971
min         1.000000
25%         1.001192
50%         1.003283
75%         1.007976
max         1.104767
Name: high_close_ratio, dtype: float64
```

In [57]:
```
# df_t=8 GARCH. Simulated data. Clipped. 'high_close_ratio'
df_clipped_sim_new["high_close_ratio"].describe()
```

Out[57]:
```
count    6468.000000
mean        1.005811
std         0.007442
min         1.000000
25%         1.001175
50%         1.003184
75%         1.007664
max         1.094833
Name: high_close_ratio, dtype: float64
```

In [58]:
```
# Real data 'fwd_return'
vet_features_df[selected_factors_list + ["hmm_regime"] + ['fwd_return']]["fwd_return"].describe()
```

Out[58]:
```
count    6468.000000
mean        0.000312
std         0.012166
min        -0.119841
25%        -0.004704
50%         0.000645
75%         0.005860
max         0.115800
Name: fwd_return, dtype: float64
```

In [59]:
```
# df_t=8 GARCH. Simulated data. Clipped. 'fwd_return' Destribution 0.02
df_clipped_sim_new["sim_return"].describe()
```

Out[59]:
```
count    6468.000000
mean        0.000217
std         0.014901
min        -0.126992
25%        -0.005982
50%         0.000504
75%         0.006964
max         0.101665
Name: sim_return, dtype: float64
```

Autocorrelation

In [60]:
```
# Real data
vet_features_df['fwd_return'].autocorr(1)
```

Out[60]:
```
np.float64(-0.10397140272057682)
```

In [61]:
```
# df_t=8 GARCH. Simulated clipped  a=0.02 b=0.90
df_clipped_sim_new['sim_return'].autocorr(1)
```

Out[61]:
```
np.float64(-0.10722737759210585)
```

Regime-wise volatility match

In [62]:
```
# Real Regime Vol
vet_features_df.groupby('hmm_regime')['fwd_return'].std()
```

Out[62]:
```
hmm_regime
0    0.005932
1    0.009972
2    0.021721
Name: fwd_return, dtype: float64
```

In [63]:
```
# df_t=8 GARCH. Simulated Regime vol. Clipped   a=0.02 b=0.90
df_clipped_sim_new.groupby('hmm_regime')['sim_return'].std()
```

Out[63]:
```
hmm_regime
0    0.007028
1    0.012157
2    0.026531
Name: sim_return, dtype: float64
```

> Regime Volatility — Correct Shape, Slightly Over-Scaled

Tail quantile comparison

Tail Behavior

In [64]:
```
# Real World
vet_features_df['fwd_return'].quantile([0.01, 0.99])
```

Out[64]:
```
0.01   -0.03419
0.99    0.03353
Name: fwd_return, dtype: float64
```

In [65]:
```
# df_t=8 GARCH. Simulated clipped  a=0.02 b=0.90
df_clipped_sim_new['sim_return'].quantile([0.01, 0.99])
```

Out[65]:
```
0.01   -0.046061
0.99    0.043582
Name: sim_return, dtype: float64
```

> Slightly Too Fat (too aggressive in heavy tails)

Factor Distribution

In [66]:
```
# Real world
vet_features_df['VIX_indx_close'].quantile([0.01, 0.99])
```

Out[66]:
```
0.01    10.020000
0.99    53.016201
Name: VIX_indx_close, dtype: float64
```

In [67]:
```
# df_t=8 GR Simulated clipped
df_clipped_sim_new['VIX_indx_close'].quantile([0.01, 0.99])
```

Out[67]:
```
0.01     9.9000
0.99    51.3465
Name: VIX_indx_close, dtype: float64
```

That is extremely close. This confirms:

> Copula marginals are correct

> Clipping barely activates

> No distortion after simulation

Correlation check

```
If that is near zero -> good (independent world).
If it’s high -> you accidentally leaked information.
```

In [68]:
```
# df_t=8 GARCH a=0.02 b=0.90
sim_return = df_clipped_sim_new['sim_return']
real_return = vet_features_df['fwd_return']

correlation_value = sim_return.corr(real_return.reset_index(drop=True))
print(f"The correlation between df_clipped_sim_new['sim_return'] and vet_features_df['fwd_return'] is: {correlation_value}")

# Calculate the correlation matrix
correlation_matrix = np.corrcoef(sim_return.abs(), real_return.abs())
print(f"The correlation_matrix is:\n {correlation_matrix}")
```

```
The correlation between df_clipped_sim_new['sim_return'] and vet_features_df['fwd_return'] is: 0.003838013469226042
The correlation_matrix is:
 [[ 1.         -0.02628604]
 [-0.02628604  1.        ]]
```

The correlation is near zero -> good, we have independent world

```
~0 means: simulator generates structurally similar but independent universes.
```

Perform the Ljung–Box test

```
Purpose:
Test whether the time series has significant autocorrelation up to a given lag.
```

In [69]:
```python
# Real World
acorr_ljungbox(vet_features_df['fwd_return'], lags=[5,10,20])
```

Out[69]:

	lb_stat	lb_pvalue
5	78.754034	1.529125e-15
10	107.486566	1.712345e-18
20	171.892935	3.712023e-26

In [70]:
```python
"""
Very small p-values ->
Strong rejection of “no autocorrelation”.
real returns have a significant sequential structure.
"""
```

Out[70]:
```
'\nVery small p-values ->\nStrong rejection of “no autocorrelation”.\nreal returns have a significant sequential structure.\n'
```

In [71]:
```python
# df_t=8 GARCH. Simulated clipped a=0.02 b=0.90
acorr_ljungbox(df_clipped_sim_new['sim_return'], lags=[5,10,20])
```

Out[71]:

	lb_stat	lb_pvalue
5	78.586101	1.657882e-15
10	84.485884	6.591865e-14
20	102.168706	5.143481e-13

In [ ]:
```python
"""
Simulated returns behave like the real series across multiple lags.
```

```
Also extremely small -> Also strong serial dependence.
Simulator captures:
    - Short memory well
    - Medium memory partially
    - Long memory less strongly
"""
```

Generate Simulated World

In [72]:
```
#==============================================================================
def generate_world(seed,
                   vet_features_df,
                   selected_factors_list,
                   betas,
                   factor_means,
                   return_means,
                   phi_dict,
                   sigma_dict,
                   trans_mat,
                   models=models,
                   df_t=8,
                   alpha=0.02,
                   beta_g=0.90):
    """
    Generate one complete synthetic world.
    All structure is fixed — only the random seed changes.
    """
    np.random.seed(seed)

    T = len(vet_features_df)

    # 1. Simulate regime path
    sim_regimes = simulate_regimes(T, trans_mat, init_state=0)

    # 2. Count regimes for copula sampling
    count_0 = (sim_regimes['hmm_regime'] == 0).sum()
    count_1 = (sim_regimes['hmm_regime'] == 1).sum()
    count_2 = (sim_regimes['hmm_regime'] == 2).sum()

    # 3. Fit copula models (on real data — fixed)
   # models = fit_copula(vet_features_df, selected_factors_list) - out of the loop now

    # 4. Sample factors per regime
    sim_F0 = sample_copula(models, selected_factors_list, regime=0, n=count_0)
    sim_F1 = sample_copula(models, selected_factors_list, regime=1, n=count_1)
    sim_F2 = sample_copula(models, selected_factors_list, regime=2, n=count_2)
```

```
    # 5. Assemble factor DataFrame in regime order
    regime_positions = {0: 0, 1: 0, 2: 0}
    rows_list = []
    for regime in sim_regimes['hmm_regime']:
        regime = int(regime)
        pos = regime_positions[regime]
        if regime == 0:   row = sim_F0.iloc[[pos]]
        elif regime == 1: row = sim_F1.iloc[[pos]]
        else:             row = sim_F2.iloc[[pos]]
        rows_list.append(row)
        regime_positions[regime] += 1

    df_world = pd.concat(rows_list, ignore_index=True)
    df_world['hmm_regime'] = sim_regimes['hmm_regime'].values

    # 6. Clip to empirical bounds
    df_world = clip_to_empirical_by_regime(
        df_world, vet_features_df, selected_factors_list
    )

    # 7. Simulate returns
    df_world = simulate_returns_regime_engine(
        df_world, selected_factors_list,
        betas, factor_means, return_means,
        phi_dict, sigma_dict,
        regime_col='hmm_regime',
        df_t=df_t
    )

    # 8. Tag world identity
    df_world['world_id'] = seed

    return df_world
#==============================================================================
```

Generate N Simulated Worlds

In [74]:
```
# ============= Generate library ==============
N_WORLDS = 50

world_library = []
# Fit once — outside the loop
models = fit_copula(vet_features_df, selected_factors_list, regime_col="hmm_regime")
for seed in range(N_WORLDS):
    world = generate_world(seed, vet_features_df, selected_factors_list,
```

```
                            betas, factor_means, return_means,
                            phi_dict_real, sigma_dict, trans_mat, models)
    world_library.append(world)

# Stack all worlds into one DataFrame if needed
all_worlds_df = pd.concat(world_library, ignore_index=True)
print(f"Library: {N_WORLDS} worlds × {len(vet_features_df)} days = {len(all_worlds_df):,} rows")
```

```
len of sub0['data']: 2431
len of sub1['data']: 2773
len of sub2['data']: 1264
Library: 50 worlds × 6468 days = 323,400 rows
```

Destribution

In [75]:
```
# Real World
vet_features_df['fwd_return'].describe()
```

Out[75]:
```
count    6468.000000
mean        0.000312
std         0.012166
min        -0.119841
25%        -0.004704
50%         0.000645
75%         0.005860
max         0.115800
Name: fwd_return, dtype: float64
```

In [76]:
```
# Simulated N Worlds
all_worlds_df['sim_return'].describe()
```

Out[76]:
```
count    323400.000000
mean          0.000286
std           0.014360
min          -0.270685
25%          -0.006237
50%           0.000328
75%           0.006841
max           0.193836
Name: sim_return, dtype: float64
```

In [77]:
```
# Real World
vet_features_df.groupby('hmm_regime')['fwd_return'].std()
```

Out[77]:
```
hmm_regime
0    0.005932
1    0.009972
2    0.021721
Name: fwd_return, dtype: float64
```

In [78]:
```
# df_t=8 GARCH. Simulated Regime vol. Clipped   a=0.02 b=0.90
all_worlds_df.groupby('hmm_regime')['sim_return'].std()
```

Out[78]:
```
hmm_regime
0    0.007074
1    0.011872
2    0.026063
Name: sim_return, dtype: float64
```

In [79]:
```
# Real World
vet_features_df['fwd_return'].quantile([0.01, 0.99])
```

Out[79]:
```
0.01   -0.03419
0.99    0.03353
Name: fwd_return, dtype: float64
```

In [80]:
```
# df_t=8 GARCH. Simulated a=0.02 b=0.90
all_worlds_df['sim_return'].quantile([0.01, 0.99])
```

Out[80]:
```
0.01   -0.041726
0.99    0.042617
Name: sim_return, dtype: float64
```

In [81]:
```
# Generate realistic price data
def close_and_regime(returns_sim, regimes_sim, start_price=1000):

    close_sim = start_price * np.exp(np.cumsum(returns_sim))
    # Convert to data frame
    regimes_sim_df = pd.DataFrame(regimes_sim)
    close_sim_df = pd.DataFrame(close_sim)

    # Align the data (make sure dates match)
    combined_sim_df = pd.concat([close_sim_df, regimes_sim_df], axis=1).dropna()
    combined_sim_df.rename(columns={'sim_return': 'sim_close'}, inplace=True)

    return combined_sim_df
```

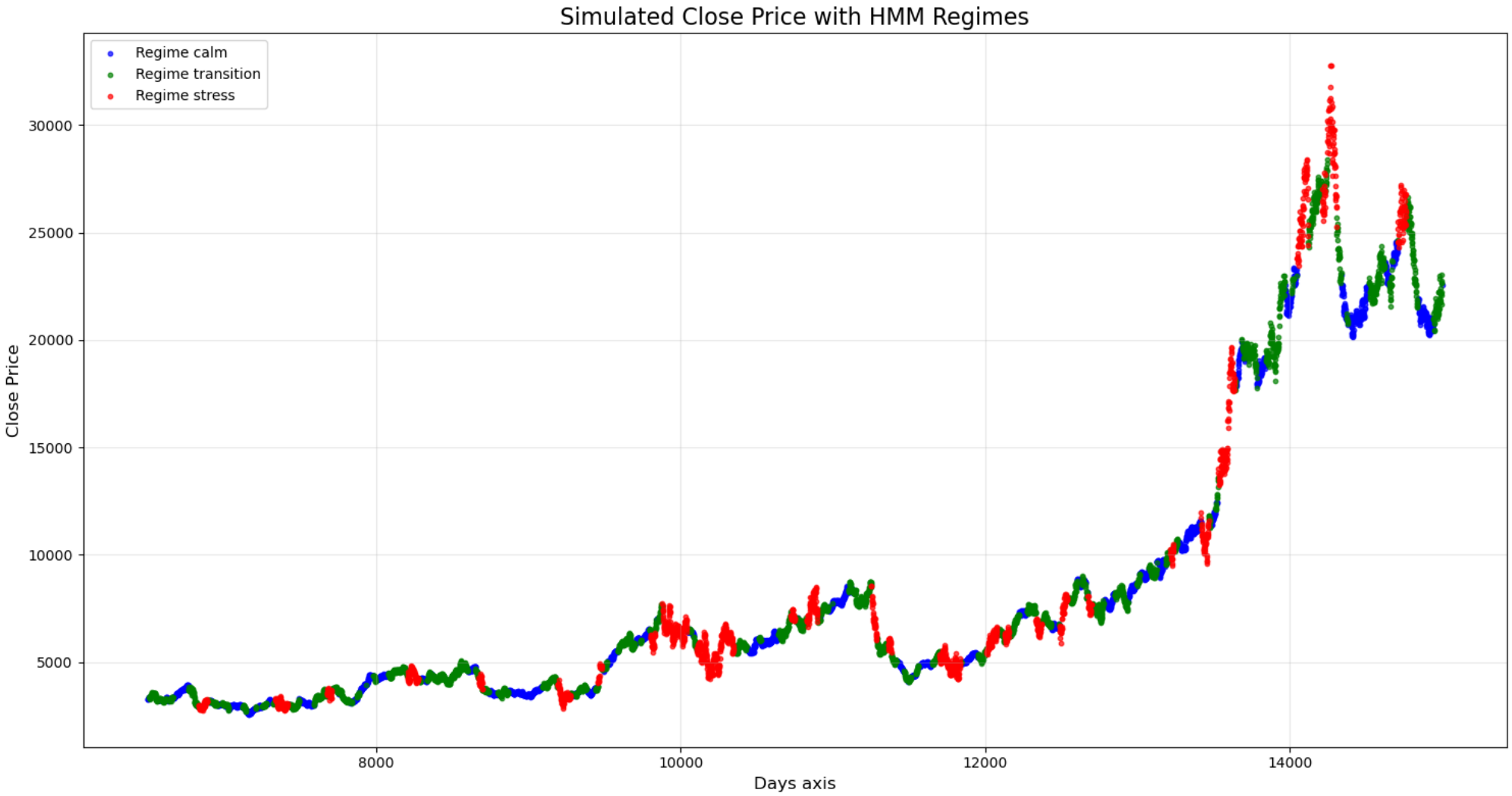
In [82]:
```
n_combined_sim_df = close_and_regime(all_worlds_df['sim_return'], all_worlds_df['hmm_regime'], start_price=1000)
```

In [84]:
```
n_combined_sim_df['hmm_regime'].value_counts()
```

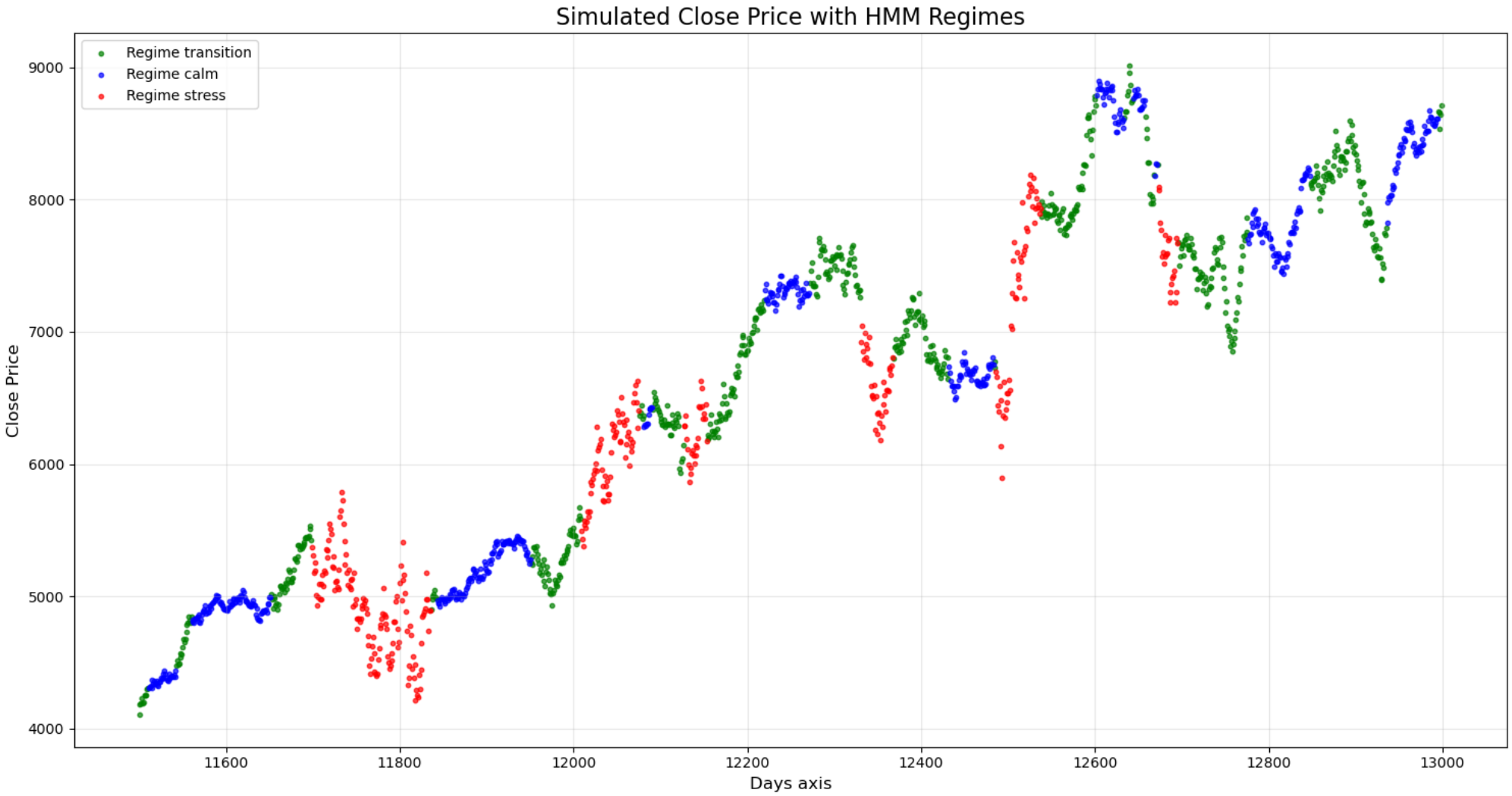
Out[84]:
```
hmm_regime
1    138358
0    124782
2     60260
Name: count, dtype: int64
```

In [87]:
```python
# Sliced vizualisation
print(f'================ Sliced vizualisation. Days 6 500 - 15 000 \n')
all_data_viz(n_combined_sim_df.iloc[6500: 15000])

# Inside sliced period
print(f'\n ================ Inside sliced period vizualisation. Days 11500 - 13000 \n')
all_data_viz(n_combined_sim_df.iloc[11500:13000])
```

```
================ Sliced vizualisation. Days 6 500 - 15 000
```




```
================ Inside sliced period vizualisation. Days 11500 - 13000
```


Simulated Close Price with HMM Regimes
Regime transition
Regime calm
Regime stress
Close Price
Days axis
9000
8000
7000
6000
5000
4000
11600
11800
12000
12200
12400
12600
12800
13000